

AMIT KUMAR

11 YOY – JAVA Full Stack

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Professional summary:

Principal Software Engineer and hands-on Java Technical Lead with 10+ years of experience building and modernizing enterprise software platforms. Strong expertise in latest Java versions, Spring Boot, REST APIs, microservices, distributed systems, event-driven architecture, concurrency, and ETL/ELT technologies. Proven experience leading Java platform and system development, modernization, designing scalable backend solutions, delivering enterprise product releases, and resolving complex production issues. Combines deep coding and system-design expertise with technical leadership, code reviews, mentoring, and delivery guidance for teams of 2–6 engineers.

Career highlights:

- **Principal Software Engineer and hands-on Java Technical Lead with 10+ years of experience** across Data Integration, FinTech, and Defence domains.
- Led the modernization of the Pentaho Java platform from **Java 8 to Java 11 and Java 17**, ensuring backward compatibility, platform stability, and support across AWS, Azure, and Google Cloud environments.
- Contributed to the development and modernization of an **AWS-hosted, microservices-based SaaS platform**, building backend APIs, enterprise data-processing capabilities, and cloud-storage integrations including Amazon S3.
- Re-architected Pentaho's legacy **OSGi-based plugin framework** into a lightweight dynamic plugin architecture using a custom Java ClassLoader, reducing platform startup time from approximately **10 minutes to 2 minutes**.
- Served as **Release Manager for Pentaho Enterprise 10.0 and 11.1**, leading branch strategy, release stabilization, quality governance, service-pack delivery, and production readiness.
- Provided hands-on technical leadership to teams of **2–6 engineers**, while actively contributing to architecture, coding, design reviews, pull-request reviews, debugging, estimation, mentoring, and production issue resolution.
- Designed and delivered enterprise-grade capabilities across **Java, Spring Boot, React, Kafka, ETL/ELT processing, database integration, Python execution, Excel processing, and cloud-storage connectors**.
- Delivered customer-driven product enhancements that improved Enterprise Edition trial-to-paid conversion by **22%**.
- Applied AI-assisted engineering practices using **GitHub Copilot, Claude, LLMs, prompt engineering, LangGraph, and workflow automation** to improve development, refactoring, documentation, code reviews, and engineering productivity.

Technical skills:

Programming languages

- Java (8/11/17)
- TypeScript
- JavaScript
- SQL
- Shell Scripting
- Angular/ React TS

Backend

- Spring Boot
- Spring MVC
- REST APIs
- Microservices
- Concurrency
- Executor Framework

Distributed Systems

- Apache Kafka
- ZooKeeper
- Redis
- Distributed Caching
- Apache Solr
- Apache Nutch

Build tools & DevOps

- Linux
- Git
- TeamCity
- UDeploy
- SonarQube
- Jenkins
- Docker

Software Architecture

- Microservices
- AWS
- Plugin Architecture
- Design Pattern
- Debugging
- Profiling

Engineering Leadership

- Architecture Reviews
- Code Reviews
- Sprint Planning
- Release Management
- Jira
- Agile scrum

Professional experiences:

Hitachi Vantara

Software Development Engineering Specialist Professional (P4) / Principal SDE

April 2022 – present

Pentaho Data Integration (PDI) is an enterprise-grade ETL/ELT and data integration platform used to design, orchestrate, and execute data pipelines across on-premises, cloud, and hybrid environments. The platform integrates with relational databases, cloud storage, streaming systems, big-data technologies, and enterprise applications.

Key Responsibilities and Achievements

- Drive architecture, hands-on development, platform modernization, and enterprise feature delivery across the Pentaho Data Integration ecosystem using **Java 17, Spring Boot, REST APIs, React, TypeScript, Kafka, AWS, and distributed-system technologies**.
- Provide hands-on technical leadership to teams of **2–6 engineers** within a larger engineering organization, while remaining actively involved in solution design, coding, pull-request reviews, debugging, estimation, mentoring, and production issue resolution.
- Led the modernization of the Pentaho platform from **Java 8 to Java 11 and Java 17**, upgrading legacy dependencies, resolving runtime compatibility issues, and maintaining backward compatibility across enterprise components.
- Coordinated platform validation and certification across supported **AWS, Microsoft Azure, and Google Cloud** environments, strengthening cloud and hyperscaler compatibility.
- Re-architected the legacy **OSGi-based plugin framework** into a lightweight dynamic plugin architecture using a custom Java ClassLoader for runtime plugin discovery and loading.
- Eliminated major OSGi runtime dependencies, simplifying deployment and maintenance while reducing application startup time from approximately **10 minutes to 2 minutes**.
- Contributed to the development and modernization of **Pipeline Designer**, an AWS-hosted, microservices-based SaaS offering for designing and executing ETL/ELT pipelines through a web-based interface.
- Designed backend services, REST APIs, business logic, and data-processing capabilities supporting scalable enterprise ETL workflows across desktop, server, and cloud deployments.
- Designed and enhanced enterprise integrations and plugins including **Kafka, Database Lookup, Combination Lookup, Dimension Lookup, Python Executor, Excel Processing, Salesforce integrations, REST Client, and cloud-storage connectors**.
- Developed cloud-storage and virtual-file-system integrations, including **Amazon S3**, enabling enterprise customers to process data across cloud and hybrid environments.
- Built reusable **React and TypeScript** components to modernize legacy desktop workflows and deliver web-based product capabilities while maintaining integration with existing Java backend services.
- Served as **Release Manager for Pentaho Enterprise 10.0 and 11.1**, managing branch strategy, branch cuts, code integration, release stabilization, defect triage, quality gates, release readiness, service-pack delivery, and production sign-off.
- Contributed to the successful delivery of Pentaho Enterprise releases **9.3, 9.4, 10.0, 10.1, 11.0, and 11.1**.
- Improved engineering quality through architecture and design reviews, unit testing, automated validation, SonarQube quality gates, CI pipelines, code reviews, and root-cause analysis of complex customer and production issues.
- Architected and implemented seamless **upgrade and downgrade support** across .exe, .tar.gz, .zip, and .jar installer distributions, eliminating the need for complete uninstall and reinstallation during product upgrades.
- Collaborated with Product Management, QA, Customer Success, DevOps, and enterprise customers to translate business requirements and customer problems into scalable product capabilities.
- Delivered customer-focused enhancements that improved Enterprise Edition product adoption and trial-to-paid conversion.
- Applied AI-assisted engineering practices using **GitHub Copilot, Claude, LLMs, prompt engineering, LangGraph, and workflow automation** to support coding, refactoring, documentation, test generation, code reviews, and engineering productivity.

Synechron Technology

Senior Associate of Technology | client: CITI

May 2021 – March 2022

- Designed and delivered a **Spring Boot–based regulatory integration solution** supporting RBI and CCIL workflows on Citi's regulatory platform.
- Collaborated with business, regulatory, and upstream/downstream technology teams to convert requirements into technical specifications, API contracts, and delivery plans.
- Developed Java services for file polling, CSV/Excel processing, REST API orchestration, JSON response handling, email notifications, Oracle audit persistence, and end-of-day reporting.
- Implemented concurrent file processing using **ExecutorService**, improving throughput while consolidating multiple input files into a single regulatory report.

- Built secure credential-rotation automation using **AES-256 encryption**, policy-based password generation, application configuration, and AutoSys scheduling.
- Designed fallback, exception-handling, and recovery mechanisms for external API integrations over publicly exposed networks.
- Created Oracle tables, SQL triggers, XML processing modules, and shell scripts supporting APAC regulatory workflows.
- Integrated CI/CD delivery with **TeamCity and UDeploy**, coordinated BAU releases, supported production deployments, and mentored three engineers.

Technology: Java 8, Spring Boot 2.5, Spring MVC, REST APIs, JPA, Oracle, Apache POI, JSON, XML/FPML, ExecutorService, AutoSys, TeamCity, UDeploy.

Bharat Electronics Limited, Ministry of Defence, Govt. Of India

Deputy Engineer | deputation: DRDO CAIR BLR

Oct 2018 – May 2021

Open-Source Intelligence Platform

Developed a distributed Open-Source Intelligence platform for collecting, processing, indexing, searching, and analysing publicly available data from web pages, social platforms, video platforms, and government sources. Collaborated with defence and intelligence stakeholders to translate analytical requirements into scalable search and data-processing capabilities.

- Designed and developed **Spring Boot–based microservices** for data collection, processing, indexing, search, and user-request management.
- Built asynchronous and event-driven communication between services using **Apache Kafka**, replacing tightly coupled synchronous integrations and improving scalability and fault isolation.
- Implemented resilient service capabilities including exception handling, failover, health monitoring, logging, security controls, Swagger/OpenAPI documentation, and Spring Boot Actuator endpoints.
- Designed data persistence for user activity, search requests, and processing metadata using **MongoDB and Spring Data**.
- Created custom **Apache Solr schemas**, indexing pipelines, and search services using SolrJ to support high-volume, low-latency information retrieval.
- Customized **Apache Nutch** crawling pipelines to enrich collected data with geographical coordinates and location information obtained through third-party APIs.
- Developed web data-collection components using **JSoup** to discover and prepare source URLs from multiple search engines and publicly available information sources.
- Designed API contracts and business workflows for communication across independently deployable microservices.
- Built and configured a distributed Linux environment using **ZooKeeper, Hadoop, HBase, Solr, MongoDB, NGINX, and virtual machines**, including networking, service discovery, load balancing, deployment, and cluster-level configuration.
- Configured **NGINX as a load balancer and reverse proxy** and deployed Java services across distributed application nodes.
- Automated environment provisioning and software installation using **Shell scripts**, reducing manual setup effort across newly created virtual machines.
- Led a team of **five engineers**, providing technical guidance, task planning, code reviews, debugging support, integration coordination, and delivery ownership.
- Collaborated with defence intelligence agencies and domain stakeholders to convert operational requirements into search capabilities, analytical workflows, and dashboard-ready datasets.

Technology: Java 8, Spring Boot, Spring MVC, Spring Data, REST APIs, microservices, Apache Kafka, Apache Solr, Indexing, Apache Nutch, MongoDB, ZooKeeper, Hadoop, HBase, JSoup, NGINX, Linux, Shell Scripting, Swagger/OpenAPI, Spring Boot Actuator, Search engine platform, Distributed systems, Shell, Linux, Angular

Fidelity Investments

Software Engineer

Aug 2015 - Sep 2018

Technology: Java, SQL, PLSQL, DB2, Oracle, COBOL, Easytrieve, VSAM, Batch Processing, Flat Files, CSV, Microsoft Excel, Microsoft Access, Spring MVC, JPA, spring transaction, multi-threading, Concurrency, Data extraction

Educational qualification:

Bachelor of Technology: Computer Science & Engineering 2010-2014

Arya College of Engineering and Research Centre, Jaipur, Rajasthan,
Rajasthan Technical University (RTU, Kota)

10th & 12th: Kendriya Vidyalaya, Vayusena Nagar, Nagpur, Maharashtra

Special certification:

MBA – Online 2022-2024

specialization: Project Management

D.Y Patil Vidhyapeeth, Pimpri, Pune MH